Luke Buchanan

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Education

Rose-Hulman Institute of Technology, Terre Haute, IN — Aug 2022 – May 2026
BS in Computer Science, Minor in Cognitive Science — GPA: 3.89

Relevant Coursework:

- Data Structures, Analysis of Algorithms, Operating Systems, Computer Architecture, Discrete Math
- Software Design, Software Requirements, Malware Analysis & Reverse Engineering, Diff Equations

Experience

Crown Equipment Corporation, New Bremen, OH — March 2025 – Aug 2025
Embedded Software Engineering Co-op

- Developed a TLS-secured approach to transmitting diagnostic data from an automation controller to a cloud-based vehicle monitoring service
- Established a custom unit testing workflow for an unsupported PLC development environment and led its adoption on a team of 5
- Collaborated with an international project team to define requirements and conduct technical investigations for a next-generation AGV platform

Skills

Languages: C, C++, Python, Java, Go, Rust, x86-64, ARM, RISC-V, Verilog, SQL, JavaScript, Racket
Software: Linux, Git, GCC, GNU Make, CMake, GDB, VectorCAST, Raylib, Pandas, Atlassian
Concepts: Communication Protocols, Microcontrollers, FPGAs, PLCs, RTOS, Agile, TDD, CI/CD, REST

Projects

Nintendo Switch Modding Library — Sept 2025 – April 2026

- Designed and implemented a rust-based mod development framework for the Nintendo Switch
- Led design on a build tool to automate incremental compilation and dependency management
- Collaborated with a team of four to find technical solutions to client-driven requirements

2D Rigid Body Physics Engine – github.com/lwbuchanan/Physics2D — July 2024 – Aug 2025

- Simulated Newtonian mechanics with dynamic tickspeed and adjustable numerical integration
- Architected a modular library compatible with multiple rendering systems and game engines
- Implemented accurate collision detection supporting arbitrary convex geometry

Stack-based RISC Processor — Oct 2023 – Nov 2023

- Designed a concise instruction set and specification for a stack-based microprocessor
- Implemented and tested a datapath according to specification in Verilog HDL
- Documented the decision making and design process to facilitate collaboration on a team of 4

Activities and Achievements

Rose-Hulman Coffee Club: Co-Founder and Member — Oct 2023 – Present
Rose-Hulman CSSE Department: TA and Grader — June 2024 – Nov 2024